Bi-Weekly Status Report

Project Name: Code Sentinel – A Multi-Agent Code Review Platform
Report Period: 20 Aug 2026 – 02 Sep 2026
Prepared By: Moksh Mehta (Frontend – on behalf of the team)
Date Submitted: 02/09/2026

Sprint/Milestone	Milestone 4 (System Design Document) complete; Backend Foundation work package (Weeks 6–7) started
% Complete (overall)	25%

Indicator	Current	Previous	Trend	Status
Overall Project	25%	15%	↑	On Track
Schedule	100%	100%	→	On Track
Scope	100%	100%	→	On Track
Quality	N/A (pre-code)	N/A (pre-code)	→	On Track
Team Contribution	100%	100%	→	On Track
Dependencies	100%	100%	→	On Track

Summary Notes:
The System Design Document (M4) was completed and submitted on 27 Aug, closing the fourth consecutive deliverable on or ahead of schedule and taking milestones M1–M4 to done on both the Trello board and the Gantt chart. The design period produced more than documentation: a 277-line PostgreSQL DDL script, four OpenAPI 3.1 contracts totalling roughly 1,770 lines, thirteen architecture and UML diagrams kept as Mermaid sources alongside rendered images, and fifteen UI/UX frames – all version-controlled in `/docs/design` so implementation starts from artefacts rather than prose. With the contracts frozen, the Backend Foundation work package opened on schedule: Turborepo monorepo setup began 31 Aug and the CI pipeline on 2 Sep. The project now moves from design into its first code-producing period, and the main watch item shifts from scope to build velocity, since agent development (14 Sep) depends on the gateway and shared service template being ready.

Key Accomplishments in Two Weeks

- System Design Document submitted (M4, 27 Aug, on schedule) – 27 pages covering architecture, schema, diagrams, API contracts and wireframes
- PostgreSQL schema authored as executable DDL – 9 enumerated types, identity/tenancy and review-pipeline table groups, every table traced to a BRD functional requirement
- OpenAPI 3.1 contracts frozen for gateway, orchestrator, agent and shared components – 18 endpoints across four specification files
- Thirteen diagrams produced (layered architecture, component, deployment, two ER models, three class diagrams, review state machine, five use-case sequence diagrams), each with a checked-in Mermaid source
- 14 UI/UX frames delivered at 3840×2160, covering dashboard, three-panel review, finding detail, configuration, PR check and error states

- Turborepo monorepo layout and CI pipeline work started on schedule (31 Aug), opening the Backend Foundation work package

Planned vs. Actual

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
System Design Document	Aug 27	—	Done	Submitted on schedule; schema, OpenAPI and 15 wireframes included
Monorepo (Turborepo) + shared configs	Aug 31–Sep 1	Sep 1	In Progress	Started on schedule; single `turbo build` across services is the exit criterion
CI pipeline (GitHub Actions)	Sep 2	Sep 2	In Progress	Lint/test/build on PR; blocks merge on failure
API Gateway scaffold + auth middleware	Sep 3–Sep 6	Sep 6	Not Started	Starts next period, no dependency open
Gateway routing + webhook endpoint	Sep 7–Sep 11	Sep 11	Not Started	Contract already fixed in `gateway.yaml`
PostgreSQL migrations + seed scripts	Sep 15–Sep 16	Sep 16	Not Started	DDL drafted during M4; migration wrapper pending

Project Risks, Issues, & Dependency

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Risk (R-01)	Rate limits/cost across LLM providers once agent development ramps up in September	Medium	Sumit Goyal	Provider-agnostic abstraction layer with fallback chain specified in the SDD; usage monitored from first agent build
Risk (R-02)	Parallel agent calls could exceed orchestrator timeout under load	Low	Sumit Goyal	Closed in design: `partial` is a first-class review status and `timed_out` agent runs are recorded rather than failing the graph
Risk (R-03)	Implementation drifting from the frozen schema and OpenAPI contracts once five members code in parallel	Medium	Ayush Soni	Contracts version-controlled in `/docs/design`; contract tests to run in CI from the gateway sprint onward

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Risk (R-04)	Agent services start 14 Sep and assume a working gateway and shared service template	Medium	Vatsal Kamani	Monorepo and CI pulled to 31 Aug–2 Sep; shared agent scaffold to be published before 14 Sep
Dependency (D-01)	GitHub App credentials/webhook secrets needed before Client Integrations (Week 13)	Low	Vatsal Kamani	Registration steps documented; to be actioned closer to Week 13

No open issues this period.

Project Decisions

Decision	Rationale	Impact	Decided By	Date
Partial result is a success state; per-agent time budget enforced by the orchestrator	One slow or failing agent must not block a review	`partial` added to review status; agent runs recorded on timeout	Team	Aug. 22
One shared REST contract (`POST /v1/review`) for all five agent services	Agents stay stateless and independently deployable, testable and reusable	Single agent specification instead of five; scaffold reusable across members	Sumit, Tej	Aug. 24
PostgreSQL 15+ with UUID keys and a `dedupe_key` hash on findings	Collapses the same issue reported by two agents without runtime comparison	De-duplication moves into the schema; simplifies aggregation	Ayush, Durgesh	Aug. 25
Every read scoped to one organization (tenant isolation)	Multi-tenant safety stated once at design time rather than patched per endpoint	Organization scope enforced across gateway routes and queries	Team	Aug. 25
Deterministic style fixes (ESLint, Prettier, Ruff, Black) bypass the LLM path	Formatting corrections are safe, cheap and need no model call	Cuts LLM cost and latency on the most common finding class	Nevil, Sumit	Aug. 26

Project Scope Changes

Change	Action	Reason	Schedule Impact	Approved By
No scope changes this period.				

Upcoming Work (Next Two Week's Plan)

- Complete the Turborepo monorepo and green the CI pipeline on `main` (Vatsal Kamani)
- API Gateway scaffold and JWT auth middleware, 3–6 Sep (Ayush Soni)
- Gateway routing to the orchestrator and the GitHub webhook receipt endpoint with HMAC-SHA256 verification, 7–11 Sep (Ayush Soni)
- PostgreSQL migrations and seed scripts from the design DDL, 15–16 Sep (Ayush Soni)
- Publish a shared agent service scaffold against the `POST /v1/review` contract, ahead of agent work starting 14 Sep
- Security Agent SAST rule integration begins 14 Sep (Tej Prakash Tak); Style Agent language dispatcher begins 14 Sep (Nevil Nandasana)
- Continue weekly progress reporting and keep the Trello board in sync with the Gantt chart

Team Member Contributions

Name: Sumit Goyal
Role: Orchestration – LangGraph workflow & LLM abstraction layer

Tasks Completed	Orchestrator class diagram and LangGraph StateGraph design; PR-review and apply-fix sequence diagrams; orchestrator OpenAPI contract
Tasks In Progress	LLM abstraction interface and prompt templates – design frozen, implementation starts Week 9
Contribution Type	Architecture / Project Management
Blockers	None
Hours/Capacity	52/200 hrs logged
Evidence	System Design Document (Section 4), `orchestrator.yaml`, orchestrator class and sequence diagrams

Name: Durgesh
Role: Orchestration – Aggregation, severity ranking & Vector DB integration

Tasks Completed	Identity and review-pipeline ER models; de-duplication and severity-ranking design; vector bookkeeping tables for ChromaDB traceability
Tasks In Progress	Embedding pipeline spike for the similarity lookup – research stage, build scheduled Week 12
Contribution Type	Architecture / Documentation
Blockers	None
Hours/Capacity	51/200 hrs logged
Evidence	ER diagrams, System Design Document (Section 3), `schema.sql` review

Name: Tej Prakash Tak
Role: Agent Services – Security Agent & Logic Agent

Tasks Completed	Shared agent REST contract co-authored; Security Agent and Logic Agent design sections; SAST tool evaluation concluded (Semgrep + Bandit selected)
Tasks In Progress	Seeded vulnerable-code test samples for the 14 Sep build start – preparation stage
Contribution Type	Research / Documentation
Blockers	None
Hours/Capacity	48/200 hrs logged
Evidence	`agent.yaml`, System Design Document (agent services section), tool evaluation notes

Name: Nevil Nandasana
Role: Agent Services – Style Agent

Tasks Completed	Style Agent language dispatcher and Docker sandbox design documented; on-demand review sequence diagram; deterministic-fix decision drafted
Tasks In Progress	Sandbox base image and linter wiring plan – ready for the 14 Sep start
Contribution Type	Research / Documentation
Blockers	None
Hours/Capacity	47/200 hrs logged
Evidence	System Design Document, UC-2 sequence diagram, sandbox approach notes

Name: Parin Makwana
Role: Agent Services – Performance Agent & Documentation Agent

Tasks Completed	Performance and Documentation agent design sections; review state machine diagram; confidence-threshold behaviour specified
Tasks In Progress	Sample hot-path and missing-docstring test cases – collection stage, build starts 19 Oct
Contribution Type	Documentation / Research
Blockers	None
Hours/Capacity	46/200 hrs logged
Evidence	System Design Document, review state diagram, test-case notes

Name: Ayush Soni
Role: Backend & Database – API Gateway & PostgreSQL

Tasks Completed	PostgreSQL DDL authored end to end (enumerated types, identity/tenancy and review-pipeline tables, constraints); gateway OpenAPI contract authored
Tasks In Progress	API Gateway scaffold and auth middleware – on track for the 3 Sep start
Contribution Type	Architecture / Development
Blockers	None
Hours/Capacity	52/200 hrs logged
Evidence	`schema.sql`, `gateway.yaml`, System Design Document (Sections 3 and 5)

Name: Vatsal Kamani
Role: Integrations & DevOps – Monorepo, GitHub App & Deployment

Tasks Completed	Deployment and component views for the SDD; Turborepo layout and shared configs finalized; monorepo setup started on schedule (31 Aug)
Tasks In Progress	CI pipeline (lint, test, build on PR) – started 2 Sep, targeting a green badge on `main`
Contribution Type	Architecture / Development
Blockers	None
Hours/Capacity	50/200 hrs logged
Evidence	Deployment and component diagrams, monorepo/CI plan, repository configuration

Name: Moksh Mehta
Role: Frontend – Next.js dashboard

Tasks Completed	Fifteen UI/UX frames delivered, including the design-language token sheet, three-panel review, finding detail, configuration, PR check and error states
Tasks In Progress	Component inventory mapping frames to Next.js routes – preparation for the Week 15 build
Contribution Type	Documentation / Development
Blockers	None
Hours/Capacity	50/200 hrs logged
Evidence	`/docs/design/wireframes` (15 frames), System Design Document (UI/UX section)